

FIND OUT MORE

The magic ones

By squaring numbers made of nothing but ones, you can make all the other digits appear. Even stranger, they appear in numbers that read the same forwards and backwards (palindromic numbers). The tiny twos below mean “times itself”, or “squared”.

$$1^2 = 1$$

$$11^2 = 121$$

$$111^2 = 12321$$

$$1111^2 = 1234321$$

$$11111^2 = 123454321$$

$$111111^2 = 12345654321$$

$$1111111^2 = 1234567654321$$

*Do you think this
pattern continues
forever?*



Prisoners' puzzle

Fifty prisoners are locked in cells in a dungeon. The prison guard, not realizing the doors are locked, passes each cell at bedtime and turns the key once. A second guard comes later and turns the locks in cells 2, 4, 6, 8, and so on, stopping only at multiples of 2. A third guard does the same, but stops at cells 3, 6, 9, 12, and so on, and a fourth guard turns the lock in cells 4, 8, 12, 16, and so on. This carries on until 50 guards have passed the cells and turned the locks, then all the guards go to bed. Which prisoners escape in the night?

SQUARE *and*

When you multiply a number by itself, the answer is a *square number*. We call it square because you can arrange that many objects in a square shape. The square number series is one of the most important in maths.

1

4

9

16

1



25

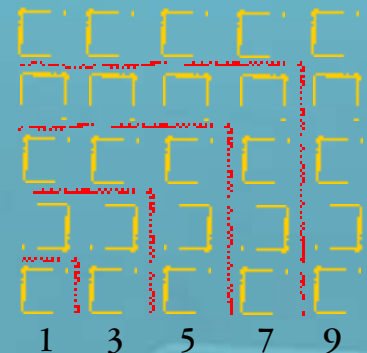


36



Something odd

The first ten square numbers are 1, 4, 9, 16, 25, 36, 49, 64, 81, and 100. Work out the difference between each pair in the sequence and write your answers down in a row. Can you spot the pattern? The diagram on the right will help you see why this pattern happens.



what comes next: 1, 4, 9, 16, 25 ..?